Arpan Pathak

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SUMMARY

Senior Software Engineer, ***8 years***: developer tools, observability platforms, and performance-critical distributed systems at ***Amazon, Microsoft, Oracle***. Full-stack ***Rust, C++, Kotlin, Go, Python, JavaScript***; ***Kubernetes***, ***eBPF/Cilium***, GPU/***CUDA***, cloud ***performance optimization***.

EXPERIENCE

Software Developer II, Microsoft, Redmond, WA — Jun 2025 – Aug 2026

- ***Cluster Observability***: ***Prometheus + Azure Monitor***, blue-green deployments, automated validation across ***AKS*** clusters in France and Germany.
- ***Kubernetes Operators & eBPF/Cilium***: Authored ***Kubebuilder operators*** reconciling network-policy state at scale; eBPF/Cilium-based L7 filtering and network visibility.
- ***C++ → Rust Migration***: Rebuilt Microsoft Information Protection from ***legacy C++/C#*** in ***Rust***, moving memory safety to compile time; ***~ 40% lower P99***, ***2x throughput***, ***3x lower memory/request***.

TechStack: ***Rust, C++, eBPF, Cilium, Kubernetes, Linux***

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA — Oct 2024 – Mar 2025

- ***Developer-Facing SDK (Go)***: Built the ***Terraform provider*** for Oracle Autonomous Database, replacing legacy control-plane tooling; ***$1.2M/month projected savings***.
- ***Database Engine Internals***: Optimized ***lock-free data structures*** in a proprietary DB engine, achieving ***+ 45,000 ops/s*** in high-concurrency read-write paths.

TechStack: ***Java, Go, Terraform, OCI***

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India — Mar 2021 – Oct 2024

- ***Real-Time ML Ranking (Python/PyTorch)***: ***XGBoost + deep-learning ensemble*** at ***100ms*** across FreeVee, Twitch, Prime Video; ***5% CTR lift***, ***$5.4M/month revenue growth***.
- ***BERT NLP Service***: ***5,000 QPS*** at peak; cut support-ticket triage SLA from ***7 days to 2 hours***.
- ***Full-Stack Developer Platforms***: ***React.js microfrontend*** migration (deploy ***5 days → 30 min***); event-bus ***SDK*** (Kotlin, ***< 50ms***) used by publisher teams.
- ***Data Platform***: ***1-petabyte*** warehouse/lake with ***Spark (AWS Glue)*** pipelines for financial reporting and BI.

TechStack: ***Kotlin, Java, Python, PyTorch, Spark, Kafka, DynamoDB, Redshift, AWS***

SDE, Razorpay, Bengaluru, India — Aug 2020 – Feb 2021

- Go UPI Payment Gateway: ***1M+ txns/hr***, ***99.99% uptime***, ***150ms P99*** at ***50K TPM***.

Software Engineer, Mindfire, Bhubaneswar, India — Aug 2018 – Feb 2020

- Java/Spring full-stack tools, AR product-search microservice (***10K+ daily queries***); latency ***5 min → 1 min***.

PERSONAL PROJECTS

Driving-CivicSense: Edge AI Vision System (Rust)

- Real-time on-device perception pipeline in ***Rust*** on the ***NVIDIA Jetson Orin Nano Super*** (67 INT8 TOPS, 8 GB, 7–15 W): ***YOLOv8/v11 INT8 ONNX*** (~ ***12 ms***) + Deep SORT/Kalman + formally-proven kinematic decision engine; ***100%*** on-device, privacy-first.
- Research paper: ***Deterministic Intersection Blockage Prediction*** | arpanpathak.github.io/driving-civicsense-vision-model/ | Code: github.com/arpanpathak/driving-civicsense-vision-model
- Books: ***Seeing Machines*** | arpanpathak.github.io/seeing-machines-book · ***CUDA Kernels*** (CUDA-Oxide / Rust-to-CUDA) | arpanpathak.github.io/gpu-parallel-book

SKILLS

Languages: Rust, C++, Kotlin, Go, C, Java, Python, TypeScript, JavaScript, Shell

Systems Programming & Performance: Lock-free data structures, concurrency, memory safety, OS internals, computer architecture, performance engineering

Networking & Protocols: TCP/IP, UDP, TLS, HTTP/2/3, QUIC, RDMA, InfiniBand, DPDK, eBPF, Cilium, VPC, NAT, VPN, load balancing

Distributed Systems & Streaming: Kafka, Flink, Spark, Spark Streaming, Kinesis, CDC, microservices, consensus/replication/sharding

Cloud & Infra: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), GCP, Kubernetes (EKS/AKS), Docker, Terraform, Kubebuilder, Istio

ML Inference & AI: PyTorch, TensorFlow, XGBoost, BERT, LLMs, ONNX Runtime, YOLOv8/v11, CNN, edge inference; CUDA, CUDA-Oxide (Rust-to-CUDA), PTX, SIMT

Observability: Prometheus, Grafana, Azure Monitor, CloudWatch, trace/sampling, profiling; DynamoDB, PostgreSQL, Redis, MongoDB, Redshift, S3, gRPC, REST, Protobuf, React.js

EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018